

Curriculum vitae

YOUNGHYUN KIM

Dr. Younghyun Kim is an Associate Professor at Hanyang University ERICA, specializing in optical communications and optical computing. His research spans silicon photonics, electronic–photonic co-integration, and photonic packaging, with a focus on co-packaged optics and photonic interconnects for next-generation AI and data-center applications.

Associate professor, Ph.D.

School of Semiconductor Convergence Engineering, BK21 FOUR ERICA-ACE Center, College of Advanced Technology and Convergence, Hanyang University ERICA

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EDUCATION

The University of Tokyo, Japan, Ph.D., Apr 2012 - Mar 2015 (일본문부성장학생)

Dept. of Electrical Engineering, Graduate School of Engineering (GPA : 3.9/4.0)

Thesis: "Study on strain-induced enhancement of plasma dispersion effect and free-carrier absorption for SiGe optical modulators/attenuators"

Advisor: Prof. Mitsuru Takenaka and Prof. Shinichi Takagi

The University of Tokyo, Japan, M.E., Apr 2010 - Mar 2012 (일본문부성장학생)

Dept. of Electrical Engineering,

Graduate School of Engineering (GPA : 3.9/4.0)

Thesis: "Research on SiGe based Modulator for Opto-Electronic Integrated Circuit"

Advisor: Prof. Mitsuru Takenaka and Prof. Shinichi Takagi

The University of Tokushima, Japan, B.E., Apr 2007 - Mar 2010 (한일공동국비유학생)

Electrical Electronic Engineering,

Faculty and School of Engineering (GPA : 5.0/5.0),

Department Rank 1, First early graduation for 6 semesters in E.E. department

Thesis: "Study on C-V Characteristics of GaN MOSFETs"

Advisor: Prof. Yasuo Ohno and Prof. Jin-Ping Ao

WORK EXPERIENCE

Hanyang University ERICA, Associate professor, Sep 2024 - Present

School of Semiconductor Convergence Engineering, College of Advanced Technology and Convergence
(첨단융합대학, 차세대반도체융합공학부)

Hanyang University ERICA, Assistant professor, Sep 2020 - Aug 2024

Department of Photonics and Nanoelectronics
Department of Applied Physics (Joint appointment)

IMEC, R&D Engineer, Jul 2018 - Aug 2020

Si Photonics team, 3D I/O group

Yonsei University, Postdoctoral Researcher, Apr 2017 - May 2018

High Speed Circuit & System Lab., Prof. Woo-young Choi
Alternative military service in South Korea (병역특례, 3-years obligation, Apr 2015 - Apr 2018)

Seoul Semiconductor/Viosys, Senior engineer, Apr 2015 - Mar 2017

Frontier R&D center
Alternative military service in South Korea (병역특례, 3-years obligation, Apr 2015 - Apr 2018)

IMEC, Belgium, Internship, Aug 2014 - Sept 2014

Si photonics team,
Advisor: Dr. Marianna Pantouvaki and Dr. Joris Van Compenhout

Samsung Advanced Institute of Technology, Korea, Internship Jul 2013 - Aug 2013

Graphene transistor team,
Advisor: Dr. Hyun Jae Song and Dr. Seongjun Park

RESEARCH INTERESTS

- **Silicon Photonics and Co-packaged Optics:** Design, fabrication, and characterization of silicon photonic devices — including Mach-Zehnder modulators, ring modulators, grating couplers, and waveguides — toward co-packaged optics and panel-level optical interconnects for next-generation AI and data-center applications. [[Optics Express 2025](#)] [[Optics Letters 2026](#)] [[Photonics Research 2022](#)]
- **Oxide Semiconductor and its Memory Applications:** Exploration of amorphous oxide semiconductor (AOS) thin-film transistors (TFTs) — including a-IGZO and a-ITZO — for ferroelectric memory (FeFET, FeDRAM) and neuromorphic display applications. [[Nanoscale 2024](#)] [[ACS AEM 2025](#)] [[IEEE TED 2024](#)]

RESEARCH GRANTS**수행중**

- 정부과제, "200 Gb/s/λ급 Tx-Rx 고집적 Co-Packaged Optics-초미세 인터커넥트 고신뢰 통합 설계·검증 플랫폼 구축", 한국산업기술기획평가원, 민관공동투자반도체고급인력양성, 과제책임자(세부), 2026.04.01 ~ 2030.12.31

- 정부과제, "2.5D 광패키징 기반 CPO를 위한 Opto-chiplet이 집적된 인터포저 기술", (재)한국연구재단, 미래유망 원천기술개발사업(차세대광패키징기술개발), 과제책임자(세부), 2025.06.01 ~ 2029.12.31
- 정부과제, "광 변조기 집적 다채널 광 배선 모듈 칩 개발", 중소기업기술정보진흥원, 창업성장기술개발사업(TIPS), 과제책임자(세부), 2024.07.01 ~ 2026.12.31
- 정부과제, "초고속/저지연 Tbps급 Optical I/O향 공진기 기반 실리콘 포토닉스 광변조기 연구", (재)한국연구재단, 이공분야기초연구사업(우수연구-신진연구), 과제책임자(총괄), 2024.04.01 ~ 2027.03.31
- 산학과제, "광패키지 탑재 미래 반도체용 저손실 광배선 기술 개발", 삼성전자(미래기술산학), 과제책임자(총괄), 2023.11.01 ~ 2026.10.31

신청중

- 정부과제, "분산형 하이브리드 양자 컴퓨팅을 위한 중성원자와 광집적회로 시스템 간의 얽힘 구현", (재)한국연구재단, 기초연구실지원사업, 과제책임자, 2026.07.01 ~ 2029.06.30
- 정부과제, "광학 인공지능경망 고속 연산 구현을 위한 대규모 병렬연산 자유공간 광학 프로세서 개발", (재)한국연구재단, 차세대광기반연산반도체핵심기술개발사업, 과제책임자(세부), 2026.06.01 ~ 2030.12.31

수행완료

- 정부과제, "차세대 디스플레이용 마이크로 광/전자소자 열분석 및 성능 향상 구조 연구", 한국기초과학지원연구원, 과제책임자(집단연구), 2023.04.01 ~ 2026.02.28
- 용역과제, "SiN 포토닉스 기반 광소자 시뮬레이션 및 설계", 한국전자통신연구원, 과제책임자, 2024.02.01 ~ 2024.07.31
- 정부과제, "150W급 가공용 레이저 다이오드 array 제조기술 개발", 한국산업기술기획평가원, 공동연구원, 2021.04.01 ~ 2024.12.31
- 정부과제, "칩온보드 기술이 적용된 상용화 수준의 실리콘 포토닉스 기반 400Gbps QSFP-DD 광트랜시버 개발", 정보통신기획평가원, 공동연구원, 2023.04.01 ~ 2026.12.31
- 정부과제, "이종집적 실리콘 집적광학 기반 III-V 광위상이동기 기술 개발", (재)한국연구재단, 과제책임자, 2022.10.01 ~ 2023.09.30 (IMEC 국제공동연구)
- 정부과제, "임계전압 보상가능 강유전성 박막트랜지스터를 이용한 마이크로 LED 픽셀 능동 회로 개발", (재)한국연구재단, 과제책임자, 2022.07.01 ~ 2023.01.31
- 정부과제, "Si집적광학기반의 고성능 KTN 광위상이동기의 수치계산 및 성능 최적화", (재)한국연구재단, 과제책임자, 2021.09.01 ~ 2022.08.31

TECHNOLOGY TRANSFER

- 기술명칭: 집적 광도파로와 광섬유를 결합하는 방법 (10-2024-0122931) 외 1건, 계약기술료: 5억원, 기업: (주)포스텍, 한양대에리카 산학협력단, 연구개발자: 김영현 교수, 계약일: 2025.09.01
- 기술명칭: 픽셀 회로, 이를 이용한 표시장치 및 이의 제조방법 (10-2022-0159364), 계약기술료: 500만원, 기업: (주)어드밴스트뷰테크놀러지, 한양대에리카 산학협력단, 연구개발자: 김영현 교수, 계약일: 2023.01.27

TEACHING

Conference

- (기술세미나 개최) 2025 첨단 광패키징 기술세미나 @한양대 ERICA (차세대지능형반도체사업단 후원) 2025.08.18 [\[Link\]](#)
- (Tutorial) 한국광학회 하계학술발표회2025, From wafer to Co-packaged optics: A tutorial on silicon photonics integration, 2025.07.09 [\[Link\]](#)

- (Tutorial) 한국광학회 동계학술발표회2024, Harmonizing Light and Silicon: The Art of Integration in Silicon Photonics, 2024.02.15 [\[Link\]](#)
- (Tutorial) 한국광학회 광전자 및 광통신 학술대회 COOC2023, 실리콘 포토닉스 - 집적공정, 2023.06.01 [\[Link\]](#)

Lecture

- (Undergraduate) Optoelectronic Devices, Optical Communication Engineering, Semiconductor Device Physics, Semiconductor Device Processing, Engineering Programming, Electromagnetics, Display Engineering
- (Graduate) Integrated Photonics, Semiconductor Advanced Packaging

ACADEMIC SERVICE

- Technical Program Committee Member (Optical Passive Devices and Modules), Optoelectronics and Communications Conference (OECC) [\[Link\]](#)
- Secretary General, Photonics Division, The Optical Society of Korea (OSK, 한국광학회) — 포토닉스 분과, 총무간사 [\[Link\]](#)
- Secretary, Optical Science Division, The Optical Society of Korea (OSK, 한국광학회) — 광과학 분과, 간사 [\[Link\]](#)
- Technical Director (기술이사), Korean Microelectronics and Packaging Society (KMEPS, 한국마이크로전자 및 패키징학회) [\[Link\]](#)

HONORS AND AWARD

Contribution Award (기여상), The Optical Society of Korea (OSK, 한국광학회) — Photonics Conference 2025, Dec. 3, 2025 [\[Link\]](#)

Contribution Award (기여상), The Optical Society of Korea (OSK, 한국광학회) — Photonics Conference 2024, Nov. 27, 2024 [\[Link\]](#)

President's Award, National Academy of Engineering of Korea (NAEK) — 2021 Campus Patent Universiade Competition (한국공학한림원회장상, 2021년도 캠퍼스 특허 유니버시아드), Nov. 24, 2021 [\[Link\]](#)

Award for Doctoral thesis in School of engineering, EE Dept., Univ. of Tokyo, March 24 2015

Japan Government Scholarship (MEXT) for Ph.D. student, April 2012 - March 2015

Japan Government Scholarship (MEXT) for M.S. student, April 2010 - Mar 2012

Early Graduation in 3 years (Department Rank 1), Tokushima Univ., March 23, 2010

Best student Awards of Nichia Co. for outstanding academic records, Tokushima Univ., June 25 2009

Best student Awards of Nichia Co. for outstanding academic records, Tokushima Univ., July 3 2008

Award for International Communication of English Ability, Tokushima Univ., July 2 2008

Best student Award for courses in liberal arts, Tokushima Univ., April 24 2008

Korea-Japan Government Joint Scholarship, March 2006 - March 2010

PUBLICATION

JOURNAL

- Simin Chen, Taewon Jin, Xinkai Sun, Gijun Ju, Zewei Wang, Jae-Hoon Han, and **Younghyun Kim***, "Unveiling Time-Dependent Dynamics in MFMS FeFETs", IEEE Journal of the Electron Devices Society (2026), Accepted

- Seokhyeon Yoon, Taewon Jin, Heeyun Jung, and **Younghyun Kim***, "Stitching-loss-tolerant silicon nitride waveguides on glass for panel-level optical interconnects", Optics Letters, Vol. 51, Iss. 7, 1903-1906 (2026) [\[Link\]](#)
- Kangseok Kim, Jonggeon Lee, and **Younghyun Kim***, "Performance Comparison of Coupled-Resonator Optical Waveguide Mach-Zehnder Modulators with III-V SIS Structures", Scientific Reports (2026), In Press
- Jonggeon Lee, Youngjoo Bae, Minkyu Kim, and **Younghyun Kim***, "A Comparative Study of High-efficiency III-V/Si SIS Ring Modulators for O-band Co-packaged Optics", Optics Communications, Vol. 609, 133106 (2026) [\[Link\]](#)
- Heeyun Jung, Taewon Jin, Seokhyeon Yoon, Kyungjin Jo, Seokyoung Shin, Seungwoo Park, and **Younghyun Kim***, "Demonstration of a SiN Grating Coupler with a Metal Reflector on a Glass Substrate", IEEE Photonics Technology Letters, Vol. 38, Iss. 5, 341-344 (2026) [\[Link\]](#)
- Yongwoo Jeon, Hong Gu Kang, Seohee Park, **Younghyun Kim**, Seong-Yong Cho, Dennis Lee, Younghoon Kim, Sung Nam Lim, Byeong Guk Jeong, Sohee Jeong, Jung Hoon Song, and Ju Young Woo, "Enhanced Carrier Mobility-Driven Performance Improvement in Colloidal Quantum Dot Solar Cells", Inorganic Chemistry Frontiers, Vol. 13, Iss. 5, 1885-1894 (2026) [\[Link\]](#)
- Taewon Jin, Seokhyeon Yoon, Kyungjin Jo, Heeyun Jung, Seokyoung Shin, Jonggeon Lee, Seungwoo Park, Byungou Lee, Wonmyoung Choi, Minchul Jang, Soo-Yong Jung, Myunghwan Kim, Jong Jin Lee, and **Younghyun Kim***, "Silicon Nitride Photonic Platform on Glass for Scalable, High-Density Optical Redistribution Layers in Panel-Level Packaging", Optics Express, Vol. 33, Iss. 24, pp. 50422-50431 (2025) [\[Link\]](#)
- Seoungmin Park, Eun A Kim, Gijun Ju, Sinwon Choi, Hyunji Jeong, Jae-Hoon Han, Dae-Hwan Ahn, Jaekyun Kim, Moonsub Shim, Heesun Yang, Seong-Yong Cho, and **Younghyun Kim***, "Integration of Quantum Dot Light-Emitting Diodes and Charge Trap Thin-Film Transistor Arrays for Memory-in-Pixel Applications", ACS Applied Materials & Interfaces (2025) [\[Link\]](#)
- Heeyun Jung, Myung-Joon Kwack, and **Younghyun Kim***, "Simulation study of a highly efficient dual-apodized a-Si/SiN grating coupler using a metal bottom reflector on a glass substrate for co-packaged optics", Japanese Journal of Applied Physics, Vol. 64 (2025) [\[Link\]](#)
- Hyunji Jeong[‡], Dae-Hwan Ahn[‡], Seoungmin Park, Gijun Ju, Jae-Hoon Han, and **Younghyun Kim***, "Visible Light Wavelength-Dependent Erasing in AOS-Based Charge Trap TFTs for Enhanced Neuromorphic Display Performance", ACS Applied Electronic Materials, Vol. 7, Iss. 12, 5463-5472 (2025) [\[Link\]](#)
- Hyeonrak Lim, Seong Kwang Kim, Seung Woo Lee, Youngkeun Park, Jaejoong Jeong, Hojin Jeong, Jinha Lim, Dae-Myeong Geum, Jaehoon Han, **Younghyun Kim**, Jaeyong Jeong, Byung Jin Cho, and Sanghyeon Kim, "Heterogeneous 3D Sequential CFET with Strain-Engineered Ge (100) Top-Channel pMOSFET on Bulk Si (100) nMOSFET", IEEE Transactions on Electron Devices, Vol. 72, Iss. 7, 3422-3428 (2025) [\[Link\]](#)
- Daehong Kim, Jeroen De Coster, Joris Van Campenhout, Yoojin Ban, Dimitrios Velenis, Huseyin Sar, Hakim Kobbi, Rafal Magdziak, and **Younghyun Kim***, "Insertion loss and polarization-dependent loss measurement improvement to enable parallel silicon photonics wafer-level testing", Optics and Lasers in Engineering, Vol. 186 (2025) [\[Link\]](#)
- Taewon Jin, Kyungjin Jo, Seokhyeon Yoon, Heeyun Jung, Seokyoung Shin, Hyunji Jeong, Kangseok Kim, and **Younghyun Kim***, "Next-Generation Semiconductor Packaging: Status of Co-Packaged Optics based on Silicon Photonics", Journal of the Microelectronics and Packaging Society, Vol. 31, no. 4, 29-36 (2024) [\[Link\]](#)

- Tae Hyeon Noh[‡], Simin Chen[‡], Hyo-Bae Kim, Taewon Jin, Seoungmin Park, Seong Ui An, Xinkai Sun, Jaekyun Kim, Jae-Hoon Han, Ji-Hoon Ahn^{*}, Dae-Hwan Ahn^{*}, and **Younghyun Kim^{*}**, "First demonstration of 2T0C-FeDRAM: a-ITZO FET and double gate a-ITZO/a-IGZO FeFET with record-long multibit retention time of > 4 bits and > 2000 sec", *Nanoscale*, 2040-3364, 2024 [\[Link\]](#)
- Seong Ui An[‡], Dae-Hwan Ahn[‡], Gijun Ju, Simin Chen, Yo Seop Ji, Jae-Hoon Han, Jaekyun Kim, and **Younghyun Kim^{*}**, "Effect of Source/Drain Electrode Materials on the Electrical Performance and Stability of Amorphous Indium-Tin-Zinc-Oxide FETs", *IEEE Transactions on Electron Devices*, Vol. 71, Iss. 9, 5422-5437 (2024) [\[Link\]](#)
- Simin Chen[‡], Dae-Hwan Ahn[‡], Taehyeon Noh, and **Younghyun Kim^{*}**, "TCAD Simulation Study of a Recessed Channel Ferroelectric-Gate Field-Effect Transistor with a Dual Ferroelectric Gate Stack for Memory Application", *Journal of the Korean Physical Society*, Vol. 85, Iss. 1, 47-55 (2024) [\[Link\]](#)
- Xinkai Sun[‡], Jae-Hoon Han[‡], Zhenyuan Xiao, Simin Chen, Taewon Jin, Taehyeon Noh, Seoungmin Park, Jaekyun Kim, Jidong Jin^{*}, and **Younghyun Kim^{*}**, "High Performance Indium-Tin-Zinc-Oxide Thin Film Transistors with Hexamethyldisilazane Passivation", *ACS Applied Electronic Materials*, Vol. 6, Iss. 4, 2442-2448 (2024) [\[Link\]](#)
- Seong Kwang Kim, Hyeong-Rak Lim, Jaejoong Jeong, Seung Woo Lee, Juhyuk Park, Joon Pyo Kim, Jaeyong Jeong, Bong Ho Kim, Seung-Yeop Ahn, Youngkeun Park, Dae-Myoung Geum, **Younghyun Kim**, Yongku Baek, Byung Jin Cho, and Sanghyeon Kim^{*}, "Heterogeneous 3D Sequential CFETs with Ge (110) Nanosheet p-FETs on Si (100) bulk n-FETs", *IEEE Transactions on Electron Devices*, Vol. 71, Iss. 1, p392-399 (2024) [\[Link\]](#)
- Daehong Kim, Jung-Tack Yang, Woo-Young Choi, and **Younghyun Kim^{*}**, "Improved Far-Field Angle in Narrow-Ridge High-Power Laser Diodes Using a Double Stripe Structure", *IEEE Photon. J.* Vol. 15, Iss. 6 (2023) [\[Link\]](#)
- Youngjoo Bae, Seong Ui An, Taewon Jin, and **Younghyun Kim^{*}**, "A PAM-4 100Gbps single-drive strained SiGe optical lumped Mach-Zehnder modulator for O-band application", *IEEE Jour. of Quant. Elec.* Vol. 59, Iss. 6 (2023) [\[Link\]](#)
- Kangseok Kim, Gijun Ju, and **Younghyun Kim^{*}**, "Numerical analysis on light extraction efficiency of a core-shell nanorod light-emitting diode", *Curr. Opt. Photon.* 7(5): 496-503 (2023) [\[Link\]](#)
- Taewon Jin, Sanghyeon Kim, Jae-Hoon Han, Dae-Hwan Ahn, Seong Ui An, Tae Hyeon Noh, Xinkai Sun, Cheol Jun Kim, Juhyuk Park, and **Younghyun Kim^{*}**, "Demonstration of programmable light intensity of a micro-LED with a Hf-based ferroelectric ITZO TFT for Mura-free displays", *Nanoscale Advances* (2023) [\[Link\]](#)
- Sung Bok Seo, Sanghee Nah, Muhammad Sajjad, Nirpendra Singh, Youngwook Shin, **Younghyun Kim**, Jaekyun Kim, and Sangwan Sim, "Ultrafast tunable broadband optical anisotropy in two-dimensional ReS₂", *Physical Review Applied*, Vol. 18, 014010 (2022) [\[Link\]](#)
- Shinick Han, **Younghyun Kim**, Donghee Son, Hyoung Won Baac, Sang Min Won, and Changhwan Shin, "Study on memory characteristics of fin-shaped feedback field effect transistor", *Semiconductor Science and Technology*, Vol. 37, 065006 (2022) [\[Link\]](#)
- **Younghyun Kim^{*}**, Didit Yudistra, Bernadette Kunert, Marina Baryshnikova, Reynald Alcotte, Cenk Ibrahim Ozdemir, Sanghyeon Kim, Sebastien Lardenois, Peter Verheyen, Joris Van Campenhout, and Marinna Pantouvaki, "Monolithic GaAs/Si V-groove depletion-type optical phase shifters integrated in a 300mm Si photonics platform", *Photonics Research*, Vol. 10, Iss. 6, pp. 1509-1516 (2022) [\[Link\]](#)
- **Younghyun Kim**, Jae-Hoon Han^{*}, Daehwan Ahn, and Sanghyeon Kim, "Heterogeneously-integrated optical phase shifters for next-generation modulators and switches on a Silicon photonics platform: A review", *Micromachines*, vol. 12, 625 (2021) [\[Link\]](#)

- **Younghyun Kim***, Taewon Jin, and Youngjoo Bae, "A comparative simulation study on lateral and L shape pn-junction phase shifters for single-drive 50 Gbps lumped Mach-Zehnder modulators", Japanese Journal of Applied Physics, vol. 60, 052002 (2021) [\[Link\]](#)
- Sanghyeon Kim, **Younghyun Kim***, Yoojin Ban, Marianna Pantouvaki, and Joris Van Campenhout, "Simulation study of a monolithic III-V/Si V-groove carrier depletion optical phase shifter", IEEE Journal of Quantum Electronics, Vol. 56, p. 6300208 (2020) [\[Link\]](#)
- Jung-Tack Yang, **Younghyun Kim**, Marzieh Pournoury, Jae-Bong Lee, Dong-Soo Bang, Tae-Kyung Kim, and Woo-Young Choi, "Influence of Emitter Width on the Performance of 975-nm (In,Ga) (As,P)/(Al,Ga)As High-power Laser Diodes", Current Optics and Photonics, Vol. 3, No. 5, pp. 445-450 (2019) [\[Link\]](#)
- Minkyu Kim, Myungjin Shin, Min-Hyeong Kim, Byung-Min Yu, **Younghyun Kim**, Yoojin Ban, Stefan Lischke, Christian Mai, Lars Zimmermann, and Woo-Young Choi, "Large-signal SPICE model for depletion-type silicon ring modulators", Photonics Research, Vol. 7, Iss. 9, pp. 948-954 (2019) [\[Link\]](#)
- **Younghyun Kim***, Youngkwan Jo, Minkyu Kim, Byung-Min Yu, Stefan Lischke, Dieter Knoll, Lars Zimmermann, and Woo-Young Choi, "Parametric Optimization for High-speed Si Micro Ring Modulators", Japanese Journal of Applied Physics, vol. 58, 062006 (2019) [\[Link\]](#)
- **Younghyun Kim***, Jung-Tack Yang, and Woo-Young Choi, "High-power broad-area laser diode performance improvement with a double pedestal structure", Japanese Journal of Applied Physics, vol. 58, 042004 (2019) [\[Link\]](#)
- Junichi Fujikata, Masataka Noguchi, **Younghyun Kim**, Jaehoon Han, Shigeki Takahashi, Takahiro Nakamura, and Mitsuru Takenaka, "High-speed and highly efficient Si optical modulator with strained SiGe layer", Applied Physics Express, vol. 11, no. 3 (2018) [\[Link\]](#)
- M. Takenaka, **Y. Kim**, J. Han, J. Kang, Y. Ikku, Y. Cheng, J. Park, M. Yoshida, S. Takashima, and S. Takagi, "Heterogeneous CMOS Photonics based on SiGe/Ge and III-V Semiconductors Integrated on Si Platform", IEEE Journal of Selected Topics in Quantum Electronics, Invited paper, Vol. 23, Iss. 3 (2017) [\[Link\]](#)
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